Lecture Scribe CSE400

L5_A_S2.pdf
PDF

Generate a lecture scribe intended to serve as faithful, exam-ready reference material for revision.

Use only the provided context consisting of the lecture slides and relevant textbook excerpts. Do not introduce any material that does not appear in the given context.

While writing the scribe, reason step by step through all definitions, notation, assumptions, theorems or results, proofs or derivations, and worked example solutions exactly as presented in the context.

Make the logical flow between concepts explicit so that a student can clearly follow the reasoning during exam preparation.

Maintain a formal academic tone and structure the scribe clearly for revision purposes.

Below is a faithful, exam-ready **Lecture Scribe for CSE400 – Lecture 5** strictly based on the provided slides and textbook excerpts, preserving all definitions, formulas, logical steps, and worked examples exactly as presented in the lecture material

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CSE400 – Fundamentals of Probability in Computing

Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

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1. Bayes' Theorem

1.1 Weighted Average of Conditional Probabilities

Let **A** and **B** be events.

Any event **A** can be expressed as:

$$A = AB \cup AB^c$$

This follows because for an outcome to belong to **A**, it must either:

• Occur in both **A and B**, or
• Occur in **A but not in B**

Since **AB** and **AB**^{c} are mutually exclusive, by Axiom 3 of probability:

$$Pr(A) = Pr(AB) + Pr(AB^c)$$

Using conditional probability:

$$Pr(A) = Pr(A \mid B)Pr(B) + Pr(A \mid B^c)[1 - Pr(B)]$$

Interpretation:

> The probability of event **A** is a **weighted average** of conditional probabilities, where the weights are the probabilities of the conditioning events.

1.2 Learning by Example — Insurance Problem

Example 3.1 (Part 1)

An insurance company classifies people into:

- Accident-prone
- Not accident-prone

Given:

$$Pr(\text{Accident} \mid \text{accident-prone}) = 0.4$$

$$Pr(\text{Accident} \mid \text{not accident-prone}) = 0.2$$

$$Pr(\text{Accident-prone}) = 0.3$$

Let:

- A_1 = policyholder has an accident within a year
- A = policyholder is accident-prone

Apply weighted average:

$$Pr(A_1) = Pr(A_1 \mid A)Pr(A) + Pr(A_1 \mid A^c)Pr(A^c)$$

$$= (0.4)(0.3) + (0.2)(0.7)$$

$$= 0.12 + 0.14 = 0.26$$

✅ **Probability of accident = 0.26**

Example 3.1 (Part 2)

Given that the policyholder **had an accident**, find the probability they are accident-prone:

$$Pr(A \mid A_1)$$

Using Bayes' formula:

$$Pr(A \mid A_1) = \frac{Pr(A)Pr(A_1 \mid A)}{Pr(A_1)}$$

$$= \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}$$

1.3 Law of Total Probability

Let:

$$B_1, B_2, \ldots, B_n$$

be **mutually exclusive events** such that:

$$\bigcup_{i=1}^{n} B_i = B$$

Since:

$$A = \bigcup_{i=1}^{n} AB_i$$

and the events AB_i are mutually exclusive:

$$Pr(A) = \sum_{i=1}^{n} Pr(AB_i)$$

Using conditional probability:

$$Pr(A) = \sum_{i=1}^{n} Pr(A \mid B_i)Pr(B_i)$$

This is called the Law of Total Probability (Formula 3.4)

1.4 Bayes' Formula (Proposition 3.1)

Using:

$$Pr(AB_i) = Pr(B_i \mid A)Pr(A)$$

we obtain:

$$Pr(B_i \mid A) = \frac{Pr(A \mid B_i)Pr(B_i)}{\sum_{j=1}^{n} Pr(A \mid B_j)Pr(B_j)}$$

Terminology:

- **Pr(B_i)** → *a priori probability*
- **Pr(B_i | A)** → *posterior probability*

1.5 Bayes Example — Colored Cards

Three cards:

Card	Sides
1	Red – Red
2	Black – Black
3	Red – Black

One card selected randomly.

Let:

- RR = all red card
- BB = all black card
- RB = mixed card
- R = upper side is red

We seek:

$$Pr(RB \mid R)$$

Using Bayes:

$$Pr(RB \mid R) = \frac{Pr(R \mid RB)Pr(RB)}{Pr(R \mid RR)Pr(RR) + Pr(R \mid RB)Pr(RB) + Pr(R \mid BB)Pr(BB)}$$

Substitute:

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)}$$

$$= \frac{1/6}{1/3 + 1/6} = \frac{1}{3}$$

✅ **Correct probability = 1/3**

Common mistake: assuming two cases are equally likely (which they are not).

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2. Random Variables

2.1 Motivation

Often we are interested not in outcomes themselves but in a **numerical function of outcomes**.

Examples:

- Sum of dice rather than each die
- Number of heads rather than full coin sequence

2.2 Definition

A **random variable** is a real-valued function defined on the sample space:

$$X : \Omega \to \mathrm{R}$$

which assigns a real number to each sample point.

The lecture restricts attention to **discrete random variables**:

- Values are finite or countably infinite
- Actual values form a discrete subset of $\mathbb{R}$

2.3 Distribution of a Random Variable

Two components:

1. Values the variable can take
2. Probabilities of those values

For any value a:

$$\{\omega \in \Omega : X(\omega) = a\}$$

is an event, abbreviated as:

$$X = a$$

The collection:

$$Pr(X = a)$$

for all values of a is called the **distribution of X**.

2.4 Visualization

The distribution can be represented by a **bar diagram**:

• x-axis → possible values
• bar height → $Pr[X = a]$

2.5 Example — Three Coin Tosses

Let:

$$Y = \text{number of heads}$$

Possible values:

$$0, 1, 2, 3$$

Probabilities:

$$P(Y = 0) = \frac{1}{8}$$

$$P(Y = 1) = \frac{3}{8}$$

$$P(Y = 2) = \frac{3}{8}$$

$$P(Y = 3) = \frac{1}{8}$$

Since Y must take one of these values:

$$1 = \sum_{i=0}^{3} P(Y = i)$$

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3. Probability Mass Function (PMF)

3.1 Discrete Random Variable

A random variable that takes **at most countably many values** is called **discrete**.

3.2 Definition of PMF

Let:

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

be the range of X.

The function:

$$p(x_k) = Pr(X = x_k)$$

is called the **Probability Mass Function (PMF)**.

Since X must take one of its values:

$$\sum_k p(x_k) = 1$$

3.3 PMF Example (Poisson-type Form)

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

First use normalization:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Find $P(X = 0)$

$$P(X = 0) = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}$$

Find $P(X > 2)$

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

✅ Key Takeaways for Exams

✔️ Probability as weighted conditional averages
✔️ Law of Total Probability decomposes events
✔️ Bayes' Formula updates prior to posterior
✔️ Random variables map outcomes → numbers
✔️ Distribution = probabilities of values
✔️ PMF sums to 1 for discrete variables

If you'd like, I can next:

📌 Convert this into **short exam notes**
📌 Create **formula sheet**
📌 Make **practice problems with solutions**
📌 Summarize into **1-page revision sheet**

Just tell me 👍